

Santiago Sosa

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Education

Drexel University | Philadelphia, PA

September 2021 – Present

Master of Science in Electrical Engineering (GPA: 3.77)

Anticipated Graduation: June 2026

Bachelor of Science in Electrical Engineering (GPA: 3.55)

Anticipated Graduation: June 2026

Bachelor of Science in Computer Engineering (GPA: 3.55)

Anticipated Graduation: June 2026

Minor: Computer Science

- BS/MS accelerated degree program

Relevant Experience

Drexel University Electrical and Computer Engineering | Philadelphia, PA

April 2022 – Present

Student Researcher

NanoPhotonics+ Laboratory (Advisor: Dr. Adam K. Fontecchio, Ph.D.)

- Fabrication of H-PDLC volume Bragg gratings with tunable refractive index modulation for NIR-selective filtering
- Tested diffraction efficiency and angular selectivity for holographic structures
- Designed hyperspectral imaging architecture spanning visible and infrared bands using lab-made holograms and 3-D printed parts
- Integrated optical filtering, focusing, and computational reconstruction pipelines
- Programming autonomous vehicle decision making using color filtering and image segmentation
- Creating H-PDLCs (Holographic Polymer Dispersed Liquid Crystals) as switchable Bragg grating structures to enhance chemical trace detection system
- Collaborating across Drexel departments for comprehensive analysis of state-of-the-art electro-optical stress monitoring and detection system. Systematic review publication under review.

Lockheed Martin | Cherry Hill, NJ

September 2023 – September 2025

Research Engineering Co-op Sr

Advanced Technology Labs (Spectral Systems Laboratory)

- Conducted research & development in sensing, tracking, and communications for space-relevant defense systems
- Developed reinforcement learning and deep learning models for object tracking in satellite-based sensing scenarios
- Innovating RF communication hardware for scalable and deployable MIMO system
- Researching and modeling watermarking techniques and waveform encryption for wireless system security and advanced communication systems
- Supporting hypersonic defense research on interaction with various weather conditions

Lockheed Martin | Moorestown, NJ

September 2022 – March 2023

Power System Engineering Co-op

Power Subsystems (Integrated Warfare Systems and Sensors – RMS)

- Collaborated as member of tiger team to optimize Space Fence radar sensitivity, and create alternate circuit capabilities
- Performed testing and analysis of transmit hardware to enhance Space Fence radar for low earth orbit object tracking
- Traveled to Space Fence radar site on Kwajalein Atoll to perform testing on transmit assembly and power hardware
- Modeling of transmit hardware assembly to compare simulated results against hardware performance

Consulting

Fontecchio Consulting | Philadelphia, PA

June 2023 – Present

Consultant

- Conducting comprehensive teardown analysis and reporting of modern nanotechnology in Apple, Samsung, and Google products for patent litigation
- Advanced material characterization through SEM-EDS and TEM-EDS analysis
- TEM sample preparation by FIBSEM cross sectioning
- FIBSEM cross section and advanced imaging for structural analysis of complex integrated circuits
- Circuit characterization using optical microscope for transistor level analysis

Honors & Awards

SuperNova Undergraduate Research Fellows Program

September 2023 - Present

- Challenges undergraduate students to build upon their STAR or undergraduate research experience
- SuperNova status earned – highest distinction

Penonni Honors College

September 2023 - Present

- Provides opportunity for STAR Scholar Research program & SuperNova Fellows Program

Dean's List

- Per quarter basis

Relevant Coursework

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|---|--|
| • Mathematical Physics (PHYS 501) | • Applied Machine Learning Engineering (ECE 612) |
| • Solid State Electronics (ECEE 520) | • Network & Transmission Media (ECEE 517) |
| • Probability & Random Variables (ECES 521) | • Analytical Methods in Systems (ECES 510) |
| • Machine Learning & AI (ECE 610) | • HPDLC Displays (ECE 697) |

Publications

- Sosa, S.; Fontecchio, A.K.; Chrysikou, E.; Atchison, J. Beyond EDA: A Systematic Review of Multimodal Sympathetic Nervous System Arousal Classification for Stress Detection. *Sensors* **2026**

Presentations

STAR Scholar Research Program

September 2022

- “Hyperspectral Imaging System Through Holographic Polymer Dispersed Liquid Crystals (H-PDLCs) Switchable Filters”
Santiago Sosa, Marquise Pullen, Tyler Hein, and Adam K. Fontecchio, STAR Poster Session, September 2022

Activities

CIRTL Learning Community

September 2025 - Present

- Cohort of PhD & Post-Doc students undertaking self-directed active learning module in preparation for future tenure-track faculty positions
- Only non-Doctoral / Post-Doctoral invited to join cohort
- Monthly meetings with tenured faculty to discuss classroom optimization and modern pedagogy

Skills

Programming: Python (TensorFlow, PyTorch, Pandas), C/C++, Embedded C, MATLAB, VHDL, C#, Arduino, R, ROS

Platforms: MacOS, Unix, Windows

Hardware: FIBSEM, SEM-EDS, TEM, CCAs, Oscilloscope, Logic analyzer, Spectrometer, Lasers, Spectrum Analyzer

Software: Cadence Virtuoso, Altium, LTSpice, AutoCAD Fusion 360, ImageJ, Tuscen, OceanView, Unity

Languages: English (Native), Spanish (Native)

Clearance: DoD Secret